Life Memory Wall — Product Documentation

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1. Vision & Idea

Life Memory Wall is a web application for capturing, preserving, and reliving life memories as beautiful visual timelines. It replaces traditional photo albums and diary apps with a **dreamlike, emotional, storybook-style experience** — designed to feel warm and nostalgic, not like a dashboard or productivity tool.

The core philosophy: **Memories are not data. They are stories. They deserve to feel like stories.**

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2. Core Concepts

2.1 Memory Spaces

A **Memory Space** is a container for memories — like a scrapbook or journal. There are two types:

Type	Description	Visibility
Personal	Private journal for one user. Only the creator can see it.	Creator only
Group	Shared space for friends, family, or teams. Multiple members contribute.	Members only

Every user can have multiple spaces. A user only sees spaces where they are an **active member**.

2.2 Memories

A **Memory** is a single event or moment on the timeline. It contains:

- **Title** — a short name (e.g., "Landing in Bangkok")
- **Date** (and optional **End Date**) — when it happened; multi-day events are supported
- **Story** — a written narrative of the event
- **Location** — where it happened (optional)
- **Tags** — categorization labels (optional)
- **Photos** — image attachments (placeholder/future feature)
- **Reactions** — emoji reactions with counts
- **Visibility** — which members can see this memory (group spaces only)
- **Creator** — who created this memory

Memories are always displayed **sorted chronologically by date** on the timeline.

2.3 Substories (Moments)

Each memory can contain **substories** — smaller moments within the larger event. For example, a "Thailand Trip" memory might have substories like "Airport chaos", "First tuk-tuk ride", "Khao San Road at midnight".

Substories have three types:

Type	Content
Text	A written story/paragraph
Photo	A single photo with caption

Photos	A photo grid (2x2) with caption

Substories are grouped by date and displayed as a flowing sub-timeline within the memory detail view.

2.4 Users

Each user has:

- Unique ID, Name, Email, Phone, Avatar
- Can own or be a member of multiple spaces
- Sees only spaces they belong to
- Has different roles across different spaces

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3. User Flows

3.1 Authentication Flow

```
                  +-------------------+
                  |   Login Screen    |
                  |                   |
                  | [Continue Google] |
                  | [Continue Email]  |
                  | [Continue Phone]  |
                  | [Quick Login x6]  |
                  +-------------------+
                         |
            +------------+------------+
            |            |            |
      Google SSO    Email Flow    Phone Flow
      (one-tap)         |            |
            |      +----+----+  +----+----+
            |      | Enter   |  | Enter   |
            |      | Email   |  | Phone   |
            |      +----+----+  +----+----+
            |           |            |
            |      +----+----+  +----+----+
            |      | Enter   |  | Enter   |
            |      | Password|  | 6-digit |
            |      +---------+  | OTP     |
            |           |       +---------+
            |           |            |
            +-----+-----+-----+-----+
                  |
            +-----+------+
            | Space      |
            | Selector   |
            +------------+
```

Google Login: Single tap, instant login.

Email Login (two-step):

1. Enter email address -> Continue
2. Enter password -> Sign In
3. Can go back to change email at step 2

Phone Login (two-step):

1. Enter phone number (with +91 prefix) -> Send OTP
2. Enter 6-digit OTP in individual input boxes
3. Auto-submits when all 6 digits filled
4. Auto-focuses next box on digit entry
5. Backspace moves to previous box
6. Resend code option available

Quick Login: 6 test user buttons at the bottom for development/testing.

3.2 Space Selection Flow

```
+-----------------------------------------+
|          "Hello, [User Name]"            |
|   "Where do you want to go today?"      |
|                                          |
|  [Pending Requests Banner - if any]      |
|                                          |
|   (0) (0) (0) (0) (0) (+)              |
|   Space bubbles    Create New            |
|                                          |
|   [Join a space with invite code]        |
+-----------------------------------------+
```

- User sees **only their spaces** (filtered by active membership)
- Each space is a **circular bubble** with emoji, title, memory count
- Group spaces show member count and role badge
- **Red badge** on spaces with pending join requests (visible to owner/admin)
- **Pending requests banner** at top if any requests await approval
- **"+" button** to create a new space
- **"Join a space"** link to enter an invite code

3.3 Space Creation Flow

1. User taps "+" button
2. Modal opens with:
   - **Name** field (text input)
   - **Emoji picker** (10 preset emojis)
   - **Type selector** (Personal / Group toggle)
3. If Group selected: shows note that a unique invite code will be generated
4. On create: space appears immediately with user as owner
5. Group spaces auto-generate a **6-character invite code**

3.4 Join Space Flow

```
User enters code -> Request sent -> Owner/Admin approves -> User gains access
```

1. User clicks "Join a space with invite code"
2. Enters 6-character code (auto-uppercased, monospace display)

3. System validates:
    - Invalid code -> Error message
    - Already a member -> Error message
    - Already requested -> Error message
    - Valid -> Creates a **pending join request**
4. Shows "Request sent! Waiting for approval." message
5. Owner/Admin of target space sees request in:
    - Red badge on space bubble
    - Pending requests banner
    - Members modal with Approve/Reject buttons
6. On approval: user becomes active member and can see the space
7. On rejection: request is removed

3.5 Timeline View Flow

```
+-------------------------------------------+
| [<- Spaces]  Space Title + Emoji   [N]    |  <- Header with back + member count
|                                           |
|    "Description of the space"             |
|    "with Member1, Member2, ..."           |
|                                           |
|         o                                 |  <- Timeline dot
|    +--------+                             |
|    | Memory |     (alternating left/right)|
|    | Card   |                             |
|    +--------+                             |
|                  o                        |
|              +--------+                   |
|              | Memory |                   |
|              | Card   |                   |
|              +--------+                   |
|         o                                 |
|    +--------+                             |
|    | Memory |                             |
|    +--------+                             |
|                                           |
|    "...and the story continues"           |
|                                           |
|  [Home] [Explore] (+) [Explore] [User]    |  <- Floating nav
+-------------------------------------------+
```

Full Mode (no memory selected):

- Alternating left/right **polaroid-style cards** along a center dashed line
- Each card shows: photo area (gradient placeholder + tag emoji), date badge, title, story excerpt, location, tags, reactions, substory count hint
- Cards animate in from left/right as they scroll into view
- Dashed timeline path animates with scroll progress
- Mobile: single column with left-aligned timeline

Split Mode (memory selected):

```
+---+-----------------------------------------+
|   |                                         |
| S | Memory Detail View (~88% width)         |
| i |                                         |
| d | Title (large serif)                     |
| e | Date (handwriting) + Location           |
| b | Story text                              |
| a |                                         |
| r | [Stories] [Photos] tabs                  |
|   |                                         |
| T | Substory timeline...                    |
| i |                                         |
| m | [+ Add a moment]                        |
| e |                                         |
| l |                                         |
| i |                                         |
| n |                                         |
| e |                                         |
+---+-----------------------------------------+
  ~12%
```

- Left sidebar (~12% width, min 180px) shows compact card list with small dots
- Right detail area takes remaining space (~88%)
- Selected card is highlighted in sidebar
- Clicking another card in sidebar switches the detail view
- Mobile: full-screen overlay for detail view

3.6 Memory Detail View

Header Section:

- Large serif title (text-4xl on desktop, text-5xl on large screens)
- Handwriting-style date with optional date range
- Location with map pin icon
- Story text in readable sans font

Two Tabs:

Stories Tab:

- Substories grouped by date
- Each date has a marker dot on a flowing timeline line
- Text stories show title + content
- Photo stories show gradient placeholder + caption
- Photo grid stories show 2x2 grid + caption
- Empty state: "No stories yet — Add moments from this memory"

Photos Tab:

- Grid of all photos from substories (2-3 columns)
- Hover reveals photo title
- Captions shown below each photo
- Empty state: "No photos yet — Photos from your stories will appear here"

Add a Moment Form:

- "Add a moment" dashed button expands into a form
- Type selector: Story / Photo / Photos (with icons)
- Title input, content/caption textarea
- Photo upload area (placeholder)
- Cancel / Add moment buttons

3.7 Memory Creation Flow

1. User taps the glowing "+" orb in floating nav
2. **Journal-page styled modal** opens with:
    - Title (serif input with underline)
    - Date picker ("When did this happen?")
    - Story textarea ("Tell the story")
    - Photo upload area (coming soon)
    - Location input (with MapPin icon)
    - Tags input (comma-separated, with Tag icon)
    - **Visibility selector** (group spaces only):
        - Eye icon toggle: "Visible to everyone" / "Visible to N selected"
        - Expands to show checkboxes for all active members
        - Each member shows name + "(you)" indicator
    - Save button with gradient
3. Memory added to the active space's timeline, sorted by date
4. Edit mode: same modal pre-filled with existing memory data

3.8 Memory Visibility (Group Spaces)

- When creating/editing a memory in a group space, user can toggle visibility
- **Default**: Visible to everyone in the space (visibleTo is empty/undefined)
- **Restricted**: Only selected members can see the memory
- The timeline filters memories using `getVisibleMemories()` — if a memory has a `visibleTo` array, only users in that array see it
- Use case: "In a group, there can be specific moments only 2-3 people are part of"

3.9 Back Navigation

Context-aware back button in the header:

- **In detail view** -> Back to timeline (label: "Timeline")
- **In timeline view** -> Back to space selector (label: "Spaces")

3.10 Member Management

Accessible via "N members" link under group space bubbles:

Members Modal shows:

- Space name + emoji
- Member count
- **Invite code** (monospace, large tracking) with:
    - Copy code button
    - Copy invite message button (copies formatted text)
- **Pending requests** section (owner/admin only):
    - Requester name, avatar initial, request date
    - Approve (green checkmark) / Reject (red X) buttons

- **Members list** sorted by role (owner -> admin -> member):
  - Avatar initial, name, "(you)" indicator, join date
  - Role badge: Owner (gold crown), Admin (teal shield), Member (text)

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4. Data Flow

4.1 State Management (Zustand)

All application state lives in a single Zustand store. No backend — everything is in-memory with mock data.

```
AppState
├── isLoggedIn: boolean
├── currentUser: User | null
├── spaces: MemorySpace[]        <- all spaces (source of truth)
├── activeSpaceId: string | null <- which space is open
│
├── login(user?) -> sets currentUser, isLoggedIn=true
├── logout() -> clears user, activeSpaceId
├── setActiveSpace(id) -> navigates to space
│
├── getActiveSpace() -> finds space by activeSpaceId
├── getVisibleSpaces() -> filters spaces where currentUser is active member
├── getVisibleMemories(space) -> filters memories by visibleTo field
│
├── addMemory(spaceId, memory) -> appends to space.memories
├── updateMemory(spaceId, memoryId, updates) -> patches memory
├── deleteMemory(spaceId, memoryId) -> removes memory
├── addReaction(spaceId, memoryId, emoji) -> increments reaction count
├── addSubstory(spaceId, memoryId, substory) -> appends to memory.substories
│
├── addSpace(space) -> creates space with user as owner + invite code
├── requestJoinByCode(code) -> validates code, creates JoinRequest
├── approveJoinRequest(spaceId, userId) -> moves from requests to members
├── rejectJoinRequest(spaceId, userId) -> removes request
├── getPendingRequests(spaceId) -> returns joinRequests array
├── getInviteCode(spaceId) -> returns invite code
├── removeMember(spaceId, userId) -> removes from membersList
└── updateMemberRole(spaceId, userId, role) -> changes member role
```

4.2 Data Flow Diagrams

Login Flow:

```
User clicks login button
  -> login({ id, name, email })
  -> set({ isLoggedIn: true, currentUser: merged })
  -> App re-renders: !isLoggedIn=false -> shows SpaceSelector
```

Space Navigation:

```
User clicks space bubble
  -> setActiveSpace(space.id)
```

```
  -> set({ activeSpaceId: id })
  -> App re-renders: activeSpaceId exists -> shows Timeline
  -> Timeline calls getActiveSpace() -> finds space
  -> Timeline calls getVisibleMemories(space) -> filters by visibleTo
  -> Renders sorted, filtered memories
```

Join Request Flow:

```
User enters invite code
  -> requestJoinByCode(code)
  -> Validates: code exists? already member? already requested?
  -> Creates JoinRequest { userId, userName, requestedAt }
  -> Appends to space.joinRequests

Owner opens members modal
  -> Sees pending requests
  -> Clicks approve
  -> approveJoinRequest(spaceId, userId)
  -> Creates SpaceMember { role: 'member', status: 'active' }
  -> Removes JoinRequest
  -> New user can now see the space in getVisibleSpaces()
```

Memory Creation Flow:

```
User fills CreateMemoryModal form
  -> handleSave(memory)
  -> addMemory(spaceId, memory) in store
  -> memory includes visibleTo[] if visibility was restricted
  -> space.memories updated, memoryCount incremented
  -> Timeline re-renders with new memory in sorted position
```

Substory Addition Flow:

```
User fills "Add a moment" form in MemoryDetail
  -> handleAddSubstory(memoryId, substory)
  -> addSubstory(spaceId, memoryId, substory) in store
  -> Appends to memory.substories (NOT main timeline)
  -> MemoryDetail re-renders with new substory
```

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5. Invite Code System

- **Format**: 6 characters from `ABCDEFGHJKLMNPQRSTUVWXYZ23456789` (no 0, O, 1, I, L to avoid confusion)
- **Generation**: On space creation, generates code and checks against all existing codes for uniqueness
- **Collision handling**: Regenerates until unique code found
- **Display**: Monospace font, large letter-spacing for readability
- **Copy options**: Copy code only, or copy formatted invite message
- **Validation**: Case-insensitive (auto-uppercased on input)

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6. Role & Permission System

Action	Owner	Admin	Member

View space & memories	Yes	Yes	Yes
Add/edit/delete own memories	Yes	Yes	Yes
See invite code	Yes	Yes	Yes
Copy invite code/message	Yes	Yes	Yes
View pending join requests	Yes	Yes	No
Approve/reject join requests	Yes	Yes	No
See pending request badges	Yes	Yes	No

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7. Mock Data Structure

7.1 Test Users (6 total)

ID	Name	Email	Phone
u1	Jagadeesh	jagadeesh@example.com	9876543210
u2	Rahul	rahul@example.com	9876543211
u3	Priya	priya@example.com	9876543212
u4	Ankit	ankit@example.com	9876543213
u5	Vikram	vikram@example.com	9876543214
u6	Sneha	sneha@example.com	9876543215

7.2 Spaces (13 total: 6 personal + 7 group)

Personal Spaces (one per user):

Space	Owner	Memories
Personal Life	Jagadeesh	3 (Luna, Dream Job, Moving)
Rahul's Diary	Rahul	2 (Open Mic, Solo Trip)
Priya's Corner	Priya	2 (Painting Sold, Adopted Biscuit)
Ankit's Log	Ankit	1 (Side Project)
Vikram's World	Vikram	1 (First Song)
Sneha's Space	Sneha	1 (50 Books)

Group Spaces:

Space	Owner	Code	Members
Thailand Trip 2025	Jagadeesh	TH25XP	Jagadeesh, Rahul, Priya, Ankit

College Gang	Vikram	CLG4EV	Vikram, Jagadeesh, Sneha, Rahul, Priya
Startup Journey	Jagadeesh	STRT9K	Jagadeesh, Ankit, Sneha
Weekend Trekkers	Rahul	TREK77	Rahul, Jagadeesh, Vikram, Ankit
The Reading Room	Sneha	READ42	Sneha, Priya, Vikram
Goa Reunion 2024	Priya	GOA24X	All 6 users
Fitness Buddies	Ankit	FIT365	Ankit, Rahul, Jagadeesh

7.3 User-to-Space Membership Matrix

User	Spaces Visible
Jagadeesh (u1)	Personal Life, Thailand Trip, College Gang, Startup Journey, Weekend Trekkers, Goa Reunion, Fitness Buddies (7)
Rahul (u2)	Rahul's Diary, Thailand Trip, College Gang, Weekend Trekkers, Goa Reunion, Fitness Buddies (6)
Priya (u3)	Priya's Corner, Thailand Trip, College Gang, Goa Reunion (4)
Ankit (u4)	Ankit's Log, Thailand Trip, Startup Journey, Weekend Trekkers, Goa Reunion (5)
Vikram (u5)	Vikram's World, College Gang, Weekend Trekkers, The Reading Room, Goa Reunion (5)
Sneha (u6)	Sneha's Space, College Gang, Startup Journey, The Reading Room, Goa Reunion (4)

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8. UI/UX Design Philosophy

8.1 Visual Identity

- **Dreamlike, warm, nostalgic** — not clinical or corporate
- **Soft animated gradient background** that shifts between lavender, peach, blue-white, and warm orange
- **Glassmorphism** for cards and overlays (frosted glass effect with backdrop blur)
- **Polaroid-style memory cards** with slight rotation and shadow
- **Floating particle canvas** with gold and lavender dots drifting slowly
- **No harsh borders** — everything uses rounded corners (rounded-2xl, rounded-3xl, rounded-full)

8.2 Typography

- **Playfair Display** (serif) — for titles, headings, names — emotional and elegant
- **DM Sans** (sans-serif) — for body text, labels, metadata — clean and readable
- **Caveat** (handwriting) — for dates, descriptions, hints — personal and warm

8.3 Color Palette

Color	Hex	Usage
Lavender	#f0e6ff	Background accents, card gradients
Peach	#ffe8d6	Background accents, warm highlights
Gold	#d4a574	Primary accent, timeline dots, buttons, glowing orb

Coral	#e8927c	Secondary accent, delete actions, urgent badges
Teal	#7cb5a8	Success states, admin badges, approve buttons
Warm White	#fffbf5	Card backgrounds, page backgrounds
Warm Dark	#4a3728	Primary text color

8.4 Animations

- **Page transitions**: Fade-in with Framer Motion
- **Card entrance**: Slide in from left/right with spring physics
- **Floating nav**: Spring up from bottom
- **Memory detail**: Slide in from right (desktop), slide up from bottom (mobile)
- **Bubble hover**: Scale up + lift (translateY -8px)
- **Create orb**: Glow animation (pulsing box-shadow)
- **Timeline path**: Animated dashed stroke
- **Particles**: Continuous slow drift across screen

8.5 Responsive Design

- **Desktop**: Full split-view with sidebar, alternating card layout, centered timeline
- **Mobile**: Single column, left-aligned timeline, full-screen detail overlay
- Breakpoint: `md` (768px) is the primary responsive switch

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9. Floating Navigation Bar

Fixed at bottom center of screen in Timeline view:

```
[Home] [Explore] (+) [Explore] [Profile]
```

- **Home**: Returns to space selector
- **Explore**: Placeholder for future discovery feature
- **+ (Create Orb)**: Opens memory creation modal, has glowing gold-to-coral gradient with pulse animation
- **Profile**: Placeholder for future profile feature

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10. Edge Cases & Behaviors

1. **Empty space**: Shows "No memories yet" with hint to use the glowing orb
2. **Empty substories**: Shows "No stories yet — Add moments from this memory"
3. **Empty photos tab**: Shows camera icon + "No photos yet"
4. **Single-member group**: Still gets invite code, can invite others
5. **User logs in with unknown email/phone**: Falls back to default user (u1) with the entered email/phone
6. **Duplicate join request**: Shows error "You already have a pending request"
7. **Already a member**: Shows error "You are already a member of this space"
8. **Memory with no visibleTo**: Visible to all space members (default behavior)
9. **Memory with visibleTo=[]**: Also visible to all (treated same as undefined)
10. **Back button context**: Always knows whether to go to timeline or spaces based on current state

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